

AgroLink

Full-Stack Setup & Deployment Guide

Step 1: Prerequisites — Install Required Software

1.1 Java JDK 17

Check whether Java is installed and at version 17 or higher:

```
java -version
```

If not recognised or below version 17:

- Download and install **Java JDK 17** (e.g. *Eclipse Temurin* from Adoptium or Oracle JDK).
- Add **C:\Program Files\Java\jdk-17\bin** (or equivalent) to your Windows system **Environment Variable Path**.
- Set a new system variable **JAVA_HOME** pointing to **C:\Program Files\Java\jdk-17**.

1.2 Apache Maven (Build Tool)

Check version:

```
mvn -version
```

If not recognised: Download the Apache Maven ZIP from the Maven website, extract it to a directory such as **C:\maven**, and add **C:\maven\bin** to your environment **Path** variable.

1.3 Node.js & NPM (Node v18+)

Check versions:

```
node -version  
npm -version
```

If not recognised: Download and run the **Node.js LTS** installer, which automatically configures the environment paths.

1.4 MySQL Server (Version 8.x)

Check version:

```
mysql --version
```

If not recognised: Add the MySQL bin directory (typically **C:\Program Files\MySQL\MySQL Server 8.0\bin**) to your system environment **Path** variable.

Step 2: Database Setup & Seeding

2.1 Start the MySQL Service

Ensure the MySQL Server service is running on your computer.

Option A — Services panel: Press **Win + R**, type **services.msc**, locate *MySQL* or *MySQL80*, then click **Start** or **Restart**.

Option B — Command Prompt (run as Administrator):

```
net start MySQL80
```

2.2 Create the Database

Log into the MySQL command-line client:

```
mysql -u root -p
```

(Enter your MySQL root password when prompted.) Once logged in, run:

```
CREATE DATABASE agrolink_db;
```

Type **exit** and press Enter to leave the MySQL CLI monitor.

2.3 Import Schema & Seed Data

Open a Command Prompt, navigate to the root directory of your extracted AgroLink folder, and run:

```
:: Import Table Structures
mysql -u root -p agrolink_db < database\schema.sql

:: Import Mock Accounts & Seed Data
mysql -u root -p agrolink_db < database\data.sql
```

You will be prompted to enter your MySQL root password for each command.

Step 3: Configure and Run Backend (Spring Boot)

1. Open **application.properties** in a text editor or IDE.
2. Locate the database properties block (Lines 7–9) and verify your MySQL credentials:

```
spring.datasource.url=jdbc:mysql://localhost:3306/agrolink_db?useSSL=false&serverTimezone=UTC&allowPublicKeyRetrieval=true
spring.datasource.username=root
spring.datasource.password=YOUR_MYSQL_PASSWORD_HERE
```

NOTE: The default password in this configuration file is set to **Root@123**. Change it if your MySQL root password is different.

3. Open a Command Prompt in the backend directory:

```
cd backend
```

4. Clean and compile the Java application:

```
mvn clean compile
```

5. Run the Spring Boot application:

```
mvn spring-boot:run
```

The backend server will launch and bind to **http://localhost:8080**.

Step 4: Configure and Run Frontend (React Vite)

1. Open a **new** Command Prompt window and navigate to the frontend directory:

```
cd frontend
```

2. Install the necessary node modules:

```
npm install
```

3. Run the development server:

```
npm run dev
```

4. Open your browser and navigate to: **http://localhost:5173**

Step 5: Test Credentials

Once the application is running, log in using these default seeded accounts. All accounts share the password **password123**.

Role	Username / Email	Password	Purpose / Features
Farmer	farmer@agrolink.com	password123	Crop listings, cold storage slot bookings, ledger
Buyer	buyer@agrolink.com	password123	Crop shopping cart, mock Razorpay payment checkout, order tracking
Cold Storage	storage@agrolink.com	password123	Approving lease requests, setting storage metrics
Administrator	admin@agrolink.com	password123	Full admin access